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Sub Code: KCH 076
Roll No.

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B.TECH
(SEM VII)THEORY EXAMINATION 2022-23
PROCESS UTILITY & SAFETY

Time: 3 Hours

Total Marks: 100

Note: Attempt all Sections. If require any missing data; then choose suitably.

Section A

1. Attempt **all** questions in brief. **2 x 10= 20**
- (a)What are the sources of water?
 - (b) Define Refrigeration.
 - (c)Write short notes on TLV- TWA
 - (d) State the basic ingredients of a successful safety program.
 - (e)Discuss for concept of water recycling.
 - (f)Define steam economy.
 - (g) Write a short note on waste heat boilers.
 - (h) Define Material Factor.
 - (i) Define safety, risk and accident.
 - (j) How does a vacuum pump differ from a compressor?

Section B

2. Attempt any **three** of the following: **10 x 3= 30**
- (a) List various equipment's for developing vacuum in process plant? Discuss the operation of steam jet ejector for developing vacuum of higher order.
 - (b) What are the physical and chemical characteristics of water? Discuss in detail about the Ion Exchange processes.
 - (c) What is Bhopal gas tragedy and explain in detail?
 - (d) Why the steam is most preferred heating medium in chemical process plant? Explain design criteria for an efficient heating system.
 - (e) List various methods of Identifying Hazards. Write in detail about Hazard Survey.

Section C

3. Attempt any **one part** of the following: **10 x 1= 10**
- (a) Explain the concept of lime Soda process in detail.
 - (b) Discuss the Activated Sludge process with the help of neat sketch.

4. Attempt any **one part** of the following:

10 x 1= 10

- (a) Explain different type of Fire tube boilers and discuss any one of the Fire tube boiler with the help of a simple diagram.
- (b) Discuss the various types of steam traps and their important characteristics. Explain any one in detail.

5. Attempt any **one part** of the following:

10 x 1= 10

- (a) Explain in brief Fault Tree Analysis for Risk Prediction.
- (b) Explain HAZOP technique with emphasis on guidewords, parameters and possible results.

6. Attempt any **one part** of the following:

10 x 1= 10

- (a) Explain the Bath-tub curve in details.
- (b) Discuss the Safety Hierarchy and Process Safety Strategies.

7. Attempt any **one part** of the following:

10 x 1= 10

- (a) Explain the FAR and OSHA.
- (b) Write short note on automatic control systems
 - (i) interlocks and Trips
 - (ii) Alarms